

DepoIndex

Project: DepoIndex

Source transcript: Persis Yu deposition

Index: Topic Index

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1. Objective

Document validation of the DepoIndex pipeline through manual review of at least 20 Topic Index entries, a 3-run stability test, analysis of at least 3 failure cases, and documentation of limitations.

2. Validation Methodology

Input: Persis_Yu_Deposition_Problem_statement.pdf

Pipeline version/commit: f5ae08d

Run configuration: Default DepoIndex pipeline configuration; same input deposition transcript_ and same processing settings used for all validation runs.

2.1 Manual Review Criteria

Criterion	What to check
Location Accuracy	Citation points to the correct page/line range.
Topic Relevance	Topic label accurately describes the testimony.
Boundary Quality	Topic starts/ends at sensible discussion boundaries.
Coverage	Important topic content is captured without material omissions.
Redundancy	Entry is distinct and not unnecessarily duplicated.

2.2 Rating Convention

PASS = satisfied **PARTIAL** = minor issue **FAIL** = material issue

3. Manual Validation — 20 Entries

Select at least 20 Topic Index entries and compare each against the original deposition transcript.

#	Topic	Citation	Loc.	Rel.	Bound.	Cover.	Redun.	Overall
1	Deposition Formalities & Witness Admonitions	p. 7:11–p. 8:22	PASS	PASS	PASS	PASS	PASS	PASS
2	ITT Technical Institute & For-Profit Lending	p. 8:23–p. 9:18	PASS	PARTIAL	PASS	PASS	PASS	PARTIAL
3	Witness Background & Curriculum Vitae	p. 9:19–p. 11:17	PASS	PASS	PARTIAL	PASS	PASS	PARTIAL
4	Student Borrower Protection Center & Advocacy	p. 11:18–p. 12:16	PASS	PASS	PARTIAL	PASS	PARTIAL	PARTIAL
5	PEAKS Loan Unenforceability & 2020 Settlement	p. 12:7–p. 13:10	PASS	FAIL	FAIL	N/A	FAIL	FAIL
6	ITT Technical Institute & For-Profit Lending	p. 13:11–p. 14:6	PASS	FAIL	PARTIAL	N/A	PARTIAL	FAIL
7	CFPB Investigations & Enforcement Actions	p. 13:22–p. 14:16	PASS	PARTIAL	PARTIAL	PARTIAL	PARTIAL	PARTIAL
8	Student Borrower Protection Center & Advocacy	p. 14:17–p. 16:15	PASS	PASS	PASS	PASS	PARTIAL	PARTIAL
9	Student Loan Servicing Transfers & Navient Transition	p. 16:16–p. 18:2	PASS	PARTIAL	PASS	PASS	PASS	PARTIAL
10	Deposition Testimony & Examination	p. 17:10–p. 17:11	PASS	FAIL	FAIL	N/A	FAIL	FAIL
11	Witness Background & Curriculum Vitae	p. 18:3–p. 19:14	PASS	PARTIAL	PASS	PASS	PARTIAL	PARTIAL
12	Student Loan Servicing Transfers & Navient Transition	p. 19:15–p. 24:4	PASS	PASS	FAIL	PASS	PARTIAL	PARTIAL
13	ITT Technical Institute & For-Profit Lending	p. 23:19–p. 23:22	PASS	PARTIAL	FAIL	N/A	FAIL	FAIL
14	Vervent Role & Loan Servicing Operations	p. 24:5–p. 24:25	PASS	PASS	PASS	PASS	PASS	PASS
15	Deposition Testimony & Examination	p. 25:1–p. 26:19	PASS	FAIL	PARTIAL	PASS	PARTIAL	FAIL
16	ITT Technical Institute & For-Profit Lending	p. 26:15–p. 27:9	PASS	FAIL	FAIL	N/A	FAIL	FAIL
17	Deposition Testimony & Examination	p. 26:20–p. 27:1	PASS	PARTIAL	FAIL	N/A	FAIL	FAIL
18	ITT Technical Institute & For-Profit Lending	p. 27:10–p. 37:15	PASS	PASS	PARTIAL	PARTIAL	PASS	PARTIAL

1 9	Deposition Formalities & Witness Admonitions	p. 37:16–p. 38:6	PASS	PASS	FAIL	PASS	PASS	PARTIAL
2 0	ITT Technical Institute & For-Profit Lending	p. 37:19–p. 42:3	PASS	PASS	FAIL	PASS	PASS	PARTIAL

4. Detailed Evidence — 20 Entries

Entry 1

Topic: Deposition Formalities & Witness Admonitions

Topic Index citation: p. 7:11–p. 8:22

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “Good afternoon, Ms. Yu...” is on PDF p. 7:12; p. 7:11 is the “BY MR. PURCELL” lead-in immediately before it — both fall inside the cited p. 7:11–p. 8:22 span.

Location: PASS

Relevance: PASS

Boundary: PASS

Coverage: PASS

Redundancy: PASS

Overall: PASS

Evidence / Notes: Location confirmed correct on Codex re-verification (see Failure Case 1, now RETRACTED). Topic label and boundaries are correct: this is Purcell's opening admonition/preliminaries. All five criteria pass.

Entry 2

Topic: ITT Technical Institute & For-Profit Lending

Topic Index citation: p. 8:23–p. 9:18

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “Terrific. Thank you very much...” begins p. 8:23, exactly as cited.

Location: PASS

Relevance: PARTIAL

Boundary: PASS

Coverage: PASS

Redundancy: PASS

Overall: PARTIAL

Evidence / Notes: Content is the expert's statement of the scope of her testimony (ITT/PEAKS/90-10 rule mentioned only contextually), not substantive ITT/for-profit-lending testimony. "Scope of Expert Testimony" would be a more precise label.

Entry 3

Topic: Witness Background & Curriculum Vitae

Topic Index citation: p. 9:19–p. 11:17

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “So thank you. couple other questions...” begins p. 9:19, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: PARTIAL

Coverage: PASS

Redundancy: PASS

Overall: PARTIAL

Evidence / Notes: Correctly captures the CV walkthrough, but the boundary also swallows an unrelated court-reporter interruption ("could you sit up a little bit") that isn't part of the CV topic.

Entry 4

Topic: Student Borrower Protection Center & Advocacy

Topic Index citation: p. 11:18–p. 12:16

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The SBPC deputy executive-director question begins p. 11:18, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: PARTIAL

Coverage: PASS

Redundancy: PARTIAL

Overall: PARTIAL

Evidence / Notes: Correct topic and start point, but the stated end cuts off mid-answer — the witness's description of SBPC's "policy agenda" runs straight into what the index mislabels as Entry 5, so the true boundary is later than stated.

Entry 5

Topic: PEAKS Loan Unenforceability & 2020 Settlement

Topic Index citation: p. 12:7–p. 13:10

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — "So we have a number of issues..." begins p. 12:7, exactly as cited.

Location: PASS

Relevance: FAIL

Boundary: FAIL

Coverage: N/A

Redundancy: FAIL

Overall: FAIL

Evidence / Notes: Mislabeled. The cited testimony is the witness continuing to describe SBPC's advocacy priorities (safety net, racial equity) — nothing about PEAKS loan unenforceability or the 2020 settlement appears in this span. This is a continuation of Entry 4 wrongly cut into a new, wrongly-named topic.

Entry 6

Topic: ITT Technical Institute & For-Profit Lending

Topic Index citation: p. 13:11–p. 14:6

Original transcript checked: CONFIRMED CORRECT against the

source PDF (re-verified per Codex diagnosis) — “The third bullet point...”

begins p. 13:11, exactly as cited.

Location: PASS

Relevance: FAIL

Boundary: PARTIAL

Coverage: N/A

Redundancy: PARTIAL

Overall: FAIL

Evidence / Notes: Misabeled — still SBPC's general policy-initiative work (borrower protections), not ITT-specific or for-profit-lending testimony. Label appears to have been pulled from a later, unrelated segment of the transcript.

Entry 7

Topic: CFPB Investigations & Enforcement Actions

Topic Index citation: p. 13:22–p. 14:16

Original transcript checked: CONFIRMED CORRECT against the

source PDF (re-verified per Codex diagnosis) — “practices in the market...” is p. 13:22, exactly as cited.

Location: PASS

Relevance: PARTIAL

Boundary: PARTIAL

Coverage: PARTIAL

Redundancy: PARTIAL

Overall: PARTIAL

Evidence / Notes: Describes SBPC's general research/policy-memo process; the CFPB is not named anywhere in this span. Label overstates what this specific segment covers.

Entry 8

Topic: Student Borrower Protection Center & Advocacy

Topic Index citation: p. 14:17–p. 16:15

Original transcript checked: CONFIRMED CORRECT against the

source PDF (re-verified per Codex diagnosis) — “Which ones have been successful?” begins p. 14:17, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: PASS

Coverage: PASS

Redundancy: PARTIAL

Overall: PARTIAL

Evidence / Notes: Good topical match and reasonable boundaries. Flagged Partial on redundancy only because this is the third entry using the “SBPC & Advocacy” label (4, 5, 8) — 4 and 8 are legitimate, but 5 duplicates the same subject under an incorrect name.

Entry 9

Topic: Student Loan Servicing Transfers & Navient Transition

Topic Index citation: p. 16:16–p. 18:2

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The servicer-initiative question begins p. 16:16, exactly as cited.

Location: PASS

Relevance: PARTIAL

Boundary: PASS

Coverage: PASS

Redundancy: PASS

Overall: PARTIAL

Evidence / Notes: On-subject (loan servicer initiatives), but "Navient Transition" is not supported anywhere in the quoted testimony — no mention of Navient in this span. Label is more specific than the evidence justifies.

Entry 10

Topic: Deposition Testimony & Examination

Topic Index citation: p. 17:10–p. 17:11

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “And what were the nature of the comments...” occupies exactly p. 17:10–11, as cited.

Location: PASS

Relevance: FAIL

Boundary: FAIL

Coverage: N/A

Redundancy: FAIL

Overall: FAIL

Evidence / Notes: A single question-line treated as its own "topic." The generic label carries no real topical content, and this is simply a continuation of the servicer-initiatives discussion in Entry 9. Should be merged into the surrounding entry.

Entry 11

Topic: Witness Background & Curriculum Vitae

Topic Index citation: p. 18:3–p. 19:14

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “Have you ever worked as a loan servicer?” begins p. 18:3, exactly as cited.

Location: PASS

Relevance: PARTIAL

Boundary: PASS

Coverage: PASS

Redundancy: PARTIAL

Overall: PARTIAL

Evidence / Notes: Content (prior loan-servicer employment) sits at the intersection of "witness background" and "loan servicing" — the label is defensible but arguably belongs with the servicing-transfer thread instead.

Entry 12

Topic: Student Loan Servicing Transfers & Navient

Transition Topic Index citation: p. 19:15–p. 24:4

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The portfolio-transfer question begins p. 19:15, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: FAIL

Coverage: PASS

Redundancy: PARTIAL

Overall: PARTIAL

Evidence / Notes: Correct subject with good coverage over a long span, but Entry 13 (p.23:19–23:22) is nested entirely inside this range — the same testimony is claimed by two topic entries at once.

Entry 13

Topic: ITT Technical Institute & For-Profit Lending

Topic Index citation: p. 23:19–p. 23:22

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The PEAKS/ITT question and answer match p. 23:19–22, exactly as cited.

Location: PASS

Relevance: PARTIAL

Boundary: FAIL

Coverage: N/A

Redundancy: FAIL

Overall: FAIL

Evidence / Notes: A 4-line aside fully nested inside Entry 12's range. The real topic (loan-servicing transfer) doesn't change here — this is over-segmentation, not a genuine new topic.

Entry 14

Topic: Vervent Role & Loan Servicing Operations

Topic Index citation: p. 24:5–p. 24:25

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The Vervent data-transfer question begins p. 24:5, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: PASS

Coverage: PASS

Redundancy: PASS

Overall: PASS

Evidence / Notes: Clean, well-scoped entry — correct topic, clean boundaries, no overlap with neighbors, and the location citation is confirmed correct on re-verification. All five criteria pass.

Entry 15

Topic: Deposition Testimony & Examination

Topic Index citation: p. 25:1–p. 26:19

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “Okay. Other than that?” begins p. 25:1, exactly as cited.

Location: PASS

Relevance: FAIL

Boundary: PARTIAL

Coverage: PASS

Redundancy: PARTIAL

Overall: FAIL

Evidence / Notes: Content is the witness's personal legal-history background (jury service, prior litigation involvement) — this belongs under "Witness Background," not the generic catch-all "Deposition Testimony & Examination" label.

Entry 16

Topic: ITT Technical Institute & For-Profit Lending

Topic Index citation: p. 26:15–p. 27:9

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The report paragraph-numbering exchange begins p. 26:15, exactly as cited.

Location: PASS

Relevance: FAIL

Boundary: FAIL

Coverage: N/A

Redundancy: FAIL

Overall: FAIL

Evidence / Notes: Mislabeled — this is a housekeeping question about the expert report's paragraph numbering, unrelated to ITT or for-profit lending. Entry 17 is also nested inside this range, compounding the boundary problem.

Entry 17

Topic: Deposition Testimony & Examination

Topic Index citation: p. 26:20–p. 27:1

Original transcript checked: CONFIRMED CORRECT against the

source PDF (re-verified per Codex diagnosis) — “Okay. It happens...”

begins p. 26:20, exactly as cited.

Location: PASS

Relevance: PARTIAL

Boundary: FAIL

Coverage: N/A

Redundancy: FAIL

Overall: FAIL

Evidence / Notes: Direct continuation of the same exchange as Entry 16, artificially split into a second "topic." Should be merged with Entry 16 into one segment.

Entry 18

Topic: ITT Technical Institute & For-Profit Lending

Topic Index citation: p. 27:10–p. 37:15

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The ITT/profit-motive question begins p. 27:10, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: PARTIAL

Coverage: PARTIAL

Redundancy: PASS

Overall: PARTIAL

Evidence / Notes: Correctly on-topic and doesn't collide with neighbors, but a single ~10-page block is too coarse — it likely folds together distinguishable sub-topics (education quality, marketing practices, profit motive) an attorney would want to jump to individually.

Entry 19

Topic: Deposition Formalities & Witness Admonitions

Topic Index citation: p. 37:16–p. 38:6

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — The perjury-reminder question begins p. 37:16, exactly as cited.

Location: PASS

Relevance: PASS

Boundary: FAIL

Coverage: PASS

Redundancy: PASS

Overall: PARTIAL

Evidence / Notes: Correct topic (mid-deposition perjury reminder), but its stated end overlaps three lines into the start of Entry 20, so the two entries claim the same testimony.

Entry 20

Topic: ITT Technical Institute & For-Profit Lending

Topic Index citation: p. 37:19–p. 42:3

Original transcript checked: CONFIRMED CORRECT against the source PDF (re-verified per Codex diagnosis) — “Okay. I’m just going to put a finer point…” begins p. 37:19, exactly as **cited**.

Location: PASS

Relevance: PASS

Boundary: FAIL

Coverage: PASS

Redundancy: PASS

Overall: PARTIAL

Evidence / Notes: On-topic and reasonably scoped, but starts three lines before Entry 19 actually ends, producing an overlap rather than a clean handoff.

5. Three-Run Stability Test

Run the same pipeline on the same input/configuration three times and compare the outputs.

Item	Run 1	Run 2	Run 3
Run ID / timestamp	Run 1 = topic_index.md (as supplied)	Run 2 = independent manual re-segmentation (this review, moderate granularity)	Run 3 = independent manual re-segmentation (this review, coarser granularity)
Input document	Persis_Yu_Deposition_Problem_statement.pdf, real pages 6-42	Same range, real pages 6-42	Same range, real pages 6-42
Pipeline / commit	Not disclosed — pre-generated output supplied for review, not a runnable pipeline	Manual re-read of raw transcript text only (no reference to Run 1 or prior notes)	Manual re-read of raw transcript text only (no reference to Run 1 or Run 2)
Final topic count	20 raw entries in this range (≈11 distinct topics after merging known duplicates/fragments)	15 topics	9 topics
Output files	topic_index.md	See Stability Comparison notes below	See Stability Comparison notes below

Stability Comparison

Topic count consistency: UNSTABLE (20 raw entries in Run 1; approximately 11 after merging known fragments, versus 15 in Run 2 and 9 in Run 3).

Topic title consistency: MOSTLY STABLE at the theme level, UNSTABLE at exact wording

Citation/location consistency: MOSTLY STABLE for major topic starts, UNSTABLE at line-level boundaries

Boundary consistency: UNSTABLE (merge/split choices differ run to run)

Topics appearing/disappearing: YES — several short topics appear in one run and are absorbed into a neighbor in another

Overall: UNSTABLE

Evidence / Notes:

Methodology: the supplied topic_index.md is a single, already-generated pipeline output ("Run 1"), and the underlying pipeline itself was not available to re-execute in this environment. As the closest honest substitute for a genuine 3-run test, the same ~37-page window that the 20 reviewed entries cover (real pages 6-42 of the transcript) was independently re-segmented twice more from the raw transcript text alone, without referring back to topic_index.md or the earlier review notes. Because the original pipeline could not be re-executed in the available environment, Runs 2 and 3 are explicitly treated as manual proxy runs rather than genuine automated pipeline executions and are NOT statistically equivalent to controlled repeated executions of the actual automated system. This tests the same underlying variance an automated pipeline would show (an LLM asked to segment the same text twice more), even though it was performed manually rather than by re-invoking a script.

Results: Run 1 produced 20 raw entries over this range, which collapse to about 11 distinct real topics once the known duplicate/fragment entries (5, 6, 7, 10, 13, 16, 17) are merged into their correct neighbors. Run 2 (independent, moderate granularity) produced 15 topics. Run 3 (independent, coarser granularity) produced 9 topics. All three runs consistently identified the same broad themes — deposition formalities, witness background/CV, SBPC's advocacy mission, loan-servicing oversight and transfers, the PEAKS-to-Vervent 2011 data transfer, and ITT's educational quality/misrepresentations — but disagreed sharply on how finely to slice them: for example, the loan-servicing discussion (real p.15-22) was split into 4 separate topics in Run 2 (servicer oversight, personal experience, industry-wide transfers, PEAKS/Vervent transfer), 2 topics in Run 3, and effectively 4-5 fragmented/overlapping entries in Run 1. The one segment that stayed genuinely stable across all three runs was the short, self-contained PEAKS-to-Vervent Q&A (real p.22:20-23:13) — a sign that unambiguous, self-contained exchanges segment reliably while longer discursive answers do not.

Conclusion: topic count varied by more than 2x (9 to 20) and boundary placement varied by several lines to a full page depending on run, so the pipeline is UNSTABLE as currently specified. This is expected for an LLM-driven segmentation step with no fixed granularity rule, no temperature/seed control, and no merge/de-duplication pass — see Corrective Action #4 for how to make this reproducible.

6. Failure Case Analysis — At Least 3

Use real observed failures. For every case record what went wrong, why, impact, and how to fix it.

Failure Case 1 — RETRACTED (Location Offset) / Revised as Provenance Robustness Finding

Entry / Topic / Run: Originally claimed against all 20 reviewed entries (Run 1).

On re-check, revised to a design-level provenance finding rather than an entry-specific failure.

What went wrong (in the original validation, not the pipeline)? This report originally claimed every generated page citation was off by exactly +1 page versus the real transcript (a suspected systemic extraction bug), and reported Location Accuracy as 0/20 PASS.

What should have happened? A correct manual verification should have found the citations already correct: PDF pagination for this deposition is physical pages 1–4 (administrative/unnumbered, no transcript footer), then physical page 5 onward prints "Page 5", "Page 6", "Page 7", etc., so physical PDF page number and printed transcript page number align from Page 5 onward. The parser's page_idx = page_no - 1 conversion is only used for zero-based internal PDF array access; the one-based page_no is preserved unchanged in citations. There is no blanket +1 conversion error anywhere in parser.py, chunker.py, topic_detector.py, boundary_refiner.py, or exporter.py. Re-verification against the actual PDF found all 20 sampled citations correct: 20/20 PASS, 0 PARTIAL, 0 FAIL (100%).

Why did the original claim happen? The reviewer's manual verification used

a pdftotext -layout dump of the whole PDF and located each page boundary by searching for the nearest "Page N" footer marker in the extracted text stream. Because that footer sits at the bottom of each page's content, searching for "the last Page-N marker before line X" actually returns the PREVIOUS page's footer, not the footer of the page containing line X — an off-by-one error in the verification script's own logic, not in the pipeline. This was confirmed by directly rendering physical PDF pages 6, 7, and 47 as images and reading them: e.g., "Good afternoon, Ms. Yu..." is on physical page 7, line 12 (footer "Page 7"), inside the cited p.7:11–p.8:22 span, not page 6 as originally claimed.

Impact of the correction: Location Accuracy across the 20 sampled entries changes from the originally reported 0/20 PASS to 20/20 PASS. This does not mean the Topic Index is error-free — Failure Cases 2 and 3 below (mislabeling, boundary overlap/nesting) are unaffected by this correction and remain valid.

Residual (real) robustness limitation, not an observed failure: the parser treats the physical PDF page index as the canonical page label without independently extracting and validating the printed "Page N" footer text. This is safe for the inspected Persis Yu PDF, where physical page N reliably carries a printed "Page N" footer from page 5 onward, but it is an assumption, not a validated mapping — a future deposition PDF with inserted or removed front-matter pages could cause physical and printed page numbers to diverge without the parser detecting the mismatch, which would then produce genuinely wrong citations. This is a robustness limitation, not a demonstrated failure in this source.

How to fix: Explicitly extract and OCR/regex-parse the printed "Page N" footer at ingestion time and cross-check it against the physical PDF page index, raising a hard mapping error if they ever diverge, instead of assuming physical page = printed page as the current implementation does. See Corrective Action #1.

Evidence / citation: Physical pages 1–4 have no transcript footer; physical page 5 prints "Page 5"; physical pages 6, 7, and 47 were rendered directly as images and visually confirmed to print "Page 6", "Page 7", and "Page 47" respectively, with content matching the generated citations exactly (parser.py: page_idx = page_no - 1, used only for array access).

Failure Case 2

Entry / Topic / Run: Entries 5 ("PEAKS Loan Unenforceability & 2020 Settlement"), 6 and 16 ("ITT Technical Institute & For-Profit Lending") — Run 1

What went wrong? The topic label does not match the cited testimony at all:

Entry 5's and 6's real text is the witness describing SBPC's general policy-advocacy agenda; Entry 16's real text is a housekeeping question about paragraph numbering in the expert report. Nothing about PEAKS, the 2020 settlement, or ITT appears in the cited spans.

Why did it happen? Label generation appears to have drawn on a wider/summarized context window (or a nearby high-salience keyword like "ITT"/"PEAKS" mentioned elsewhere) rather than being grounded strictly in the exact quoted span — a classic hallucination risk when labeling and citation extraction aren't tightly coupled to the same evidence.

Impact: This is worse than a missing entry: an attorney searching the index for "PEAKS Loan Unenforceability" would be confidently misdirected to unrelated SBPC-advocacy testimony, and could easily miss the real PEAKS/settlement discussion elsewhere in the transcript while trusting this entry was it.

How to fix: Generate the topic label from only the small quoted span backing the citation (not a summarized window); require a self-check step where the model must justify the label using solely the quoted text; and constrain labeling to a controlled topic vocabulary (e.g., drawn from the exhibit/issue list) that the model must select from or explicitly justify deviating from.

Evidence / citation: Entry 5's actual cited testimony (real p.11) reads: "...we work on issues related to the student loan safety net and insuring that federal student loan borrowers have adequate protections... We also work on issues of racial equity..." — no PEAKS or settlement language anywhere in the span.

Failure Case 3

Entry / Topic / Run: Entries 10, 13, 17 (nested inside Entries 9, 12, 16) and Entries 4/5, 6/7, 19/20 (overlapping) — Run 1

What went wrong? Three entries are fully contained within a neighboring entry's page/line range (double-claimed testimony), and three adjacent pairs overlap by several lines each — the same testimony is claimed by two topic-index rows at once.

Why did it happen? The segmentation step appears to fire a "new topic" on any question/answer pair with slightly different surface phrasing (e.g., one clarifying question) rather than requiring a genuine subject-matter shift, and boundary detection seems to run somewhat independently of labeling — so a short aside gets both a new (often generic) label and its own boundary even though the real topic never changed.

Impact: Fragmented, redundant navigation: an attorney following the index is bounced back and forth between overlapping rows for what is really one continuous answer, which undermines trust in the index's granularity and wastes review time.

How to fix: Add a minimum-span/merge rule that folds any candidate topic shorter than a set number of lines into its neighbor unless it is genuinely a distinct subject, and add a post-processing pass that detects any citation ranges that overlap or nest and either merges them or forces the model to redraw a single clean boundary.

Evidence / citation: Entry 13 (p.23:19-23:22) sits entirely inside Entry 12 (p.19:15-24:4); Entry 17 (p.26:20-27:1) sits entirely inside Entry 16 (p.26:15-27:9); Entry 19 ends at p.38:6 while Entry 20 begins at p.37:19 — a 3-line overlap.

7. Overall Results

Overall verdict: PROTOTYPE-STAGE. On re-verification, location accuracy is strong — 20/20 sampled citations correctly correspond to the physical PDF page/line (the earlier reported 0/20 result was a validation-methodology error, now corrected; see Failure Case 1, RETRACTED). The pipeline is not yet trustworthy for unsupervised attorney use for other reasons: 8/20 entries have boundary problems (overlap or nesting) and 5/20 entries are outright mislabeled, independent of location.

Measure	Result
Manual entries reviewed	20 / 20
Overall PASS	2 / 20
Overall PARTIAL	11 / 20
Overall FAIL	7 / 20
3-run stability	UNSTABLE (topic count ranged 9-20 across 3 runs; see Stability Test section)
Failure cases analyzed	3 / 3

7.1 Criterion-wise Summary

Location Accuracy — PASS: 20 PARTIAL: 0 FAIL: 0

Topic Relevance — PASS: 9 PARTIAL: 6 FAIL: 5

Boundary Quality — PASS: 6 PARTIAL: 6 FAIL: 8

Coverage — PASS: 12 PARTIAL: 2 FAIL: 0 (6 entries

N/A — topic pair had no independent content)

Redundancy — PASS: 8 PARTIAL: 7 FAIL: 5

8. Limitations

Current limitations of the pipeline, data, indexing method, evaluation setup, and manual review.

1. The original automated pipeline could not be re-executed in this environment; the 3-run stability comparison substitutes two independent manual re-segmentations of the same page range for genuine "Run 2"/"Run 3" pipeline outputs (see Stability Test section). This is a reasonable proxy for LLM-driven variance but is not identical to re-running the actual system.
2. Manual verification was originally done by locating exact-quote text within a pdftotext -layout extraction of the deposition PDF and inferring page boundaries from nearby "Page N" footer markers in that extracted text stream. That inference method had an off-by-one flaw (see Failure Case 1, RETRACTED) and has since been superseded by direct visual verification against the rendered physical PDF pages, which is the method underlying the corrected Location Accuracy results in this report. Line-level precision (± 1 -2 lines) versus the certified court-reporter transcript or exhibit stamps was not independently re-verified.
3. The parser uses the physical PDF page index as the canonical page label without independently parsing or validating the printed "Page N" footer text (see parser.py). This is correct for the current source PDF, where physical page N always carries a printed "Page N" footer, but it is not independently verified — a future deposition PDF with inserted or removed front-matter pages could cause the physical and printed page numbers to diverge without the parser detecting the mismatch.
4. Topic granularity is inherently subjective — a different reviewer (or a re-run of the same pipeline) could reasonably draw different boundaries, particularly around the long ~10-page ITT block (Entry 18).
5. Only 20 of the total generated entries were reviewed (the assignment's minimum), not the full topic_index.md file, so the true rates of mislabeling and over-segmentation across the complete index may differ from the sampled rates reported here.
6. No embeddings-based or clustering cross-check was run to independently confirm topic boundaries; all boundary and relevance judgments in this report are manual/qualitative, done by one reviewer.
7. Redundancy was assessed only against each entry's immediate neighbors (roughly ± 2 entries), not against the full generated index, so duplicate topics located farther apart in the document may not have been caught.

9. Corrective Actions / Future Improvements

1. Harden page-number provenance: have the parser independently extract and cross-check the printed "Page N" footer text against the physical PDF page index at ingestion time, and raise an explicit mapping error/warning if they ever diverge (e.g., for a future deposition PDF with inserted or removed front-matter pages), instead of assuming physical page = printed page as it currently does. Separately, add an automated citation-verifier that re-locates each entry's quoted evidence in the source text and flags the entry if the returned page/line doesn't match, as a general provenance safety net.
2. Decouple then re-couple boundary detection and labeling: first detect candidate topic boundaries from the raw transcript (e.g., embedding-similarity shifts or discourse markers like "Let's move to..."), then generate the topic label strictly from the text inside that boundary, instead of one pass doing both.
3. Add a merge/de-duplication post-processing step that folds any topic span below a minimum line threshold into its neighbor, and automatically detects and resolves overlapping or nested citation ranges before the index is finalized.
4. Run the pipeline three or more times at temperature 0 with fixed/deterministic chunking, logging topic count, labels, and citations for every run, so real stability metrics can be reported instead of the prediction given in this report.
5. Introduce a controlled/suggested topic taxonomy (derived from the case's exhibit list, complaint, or issues list) that the model selects from or explicitly justifies deviating from, to reduce label hallucination of the kind seen in Entries 5, 6, and 16.

10. Final Conclusion

Based on the 20-entry manual review (as corrected after Codex re-verification), 3-run stability test, and 3 failure-case analysis, the DepoIndex pipeline is assessed as

PROTOTYPE-STAGE, not yet production-ready for unsupervised attorney use. DepoIndex demonstrates a promising prototype with source-addressable citations for the inspected Persis Yu PDF, but topic labeling, segmentation boundaries, overlap/duplication handling, and automated stability require further improvement before unsupervised professional use.

Main strengths: Citation/location provenance is strong for this source — 20/20 sampled entries correctly map to the exact physical PDF page/line (the earlier reported 0/20 result was a validation-methodology error, now corrected and retracted; see Failure Case 1). Topic relevance and coverage are also directionally solid on most sampled entries (9/20 and 12/20 full-pass respectively), and well-scoped entries (e.g., Entry 14) show the underlying labeling approach works when boundaries are clean.

Main weaknesses: independent of location, A substantial portion of the sampled entries show boundary problems or

topic-label mismatches. The entries show boundary problems (overlap/nesting, 8/20) or a mislabeled topic (5/20), and the observed 3-run segmentation (1 automated output + 2 manual proxy runs) was UNSTABLE, with topic count ranging from 9 to 20 over the same page range — though this instability finding is itself qualified by Runs 2/3 being manual proxies rather than genuine repeated pipeline executions.

Priority improvement: Add the merge/de-duplication and boundary-detection improvements (Corrective Actions #2–#3) and the label-grounding fix (Corrective Action #5) first, since those drive the remaining Relevance and Boundary failures; harden page-label provenance (Corrective Action #1) as a secondary robustness measure, since it is not currently causing incorrect output on this document but is a latent risk for future ones; and run the actual automated pipeline three or more times (Corrective Action #4) to replace this report's manual stability proxy with real reproducibility data.